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Semiconductor device including distributed write driving arrangement

Abstract

A semiconductor memory device includes: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWBL_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including local write drivers correspondingly coupled to the segments; and a global write driver coupled to each of the local write drivers.

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Background/Summary

PRIORITY CLAIM (1) The instant application is a continuation of U.S. application Ser. No. 17/816,048, filed Jul. 29, 2022, now U.S. Pat. No. 11,783,890, issued Oct. 10, 2023, which is a continuation application of U.S. application Ser. No. 17/241,687, filed Apr. 27, 2021, now U.S. Pat. No. 11,423,974, issued Aug. 23, 2022, which is a continuation application of U.S. application Ser. No. 16/991,449, filed Aug. 12, 2020, now U.S. Pat. No. 10,991,420, issued Apr. 27, 2021, which is a continuation application of U.S. application Ser. No. 16/503,344, filed Jul. 3, 2019, now U.S. Pat. No. 10,755,768, issued Aug. 25, 2020, which is a non-provisional application claiming priority to Provisional Application No. 62/698,517, filed Jul. 16, 2018, the entire contents of which are incorporated by reference herein.

BACKGROUND
(1) In a typical memory system, memory cells are arranged in an array. Each memory cell (also referred to as a cell) stores a datum representing one bit. Each cell is at the intersection of a row and a column. Accordingly, a particular cell is accessed by selection of the row and the column which intersect at the particular cell. Each of the cells in a column is connected to a bit line. An input/output (I/O) circuit uses the bit line to read a datum from, or write a datum to, a selected one of the bit cells in the column.
(2) Typically, there are many cells in a column. Due to varying physical distances between the I/O circuit and the cells, the bit line represents a different resistive and/or capacitive load for each of the cells in the column.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) One or more embodiments are illustrated by way of example, and not by limitation, in the figures of the accompanying drawings, wherein elements having the same reference numeral designations represent like elements throughout. The drawings are not to scale, unless otherwise disclosed.
(2) FIG. **1** is a block diagram of a semiconductor device, in accordance with some embodiments.

(3) FIG. **2** is a block diagram of an array & column driving region which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure.
(4) FIG. **3** is a circuit diagram of an array & column driving region which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure.
(5) FIGS. **4**A-**4**C are corresponding circuit diagrams of an array & column driving region which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure.
(6) FIGS. **5**A-**5**C are corresponding circuit diagrams of array & column driving regions, each of which includes a distributed write driving arrangement, in accordance with corresponding embodiments of the present disclosure.
(7) FIG. **6** is a cross-section of an array & column driving region **600** which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure.
(8) FIG. **7** is a flowchart of a method **700** of write-driving a column in an array & column driving region of a SRAM macro on a distributed basis, in accordance with some embodiments.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, materials, values, steps, operations, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(11) In some embodiments, for SRAM bit cells which include separate write and read ports, a distributed write driving arrangement is provided. More particularly, such a distributed write driving arrangement includes: a global write driver; and, in each segment of bit cells, a local write driver. The global write driver includes: a first inverter connected between a global write bit (GWB) line and a local write bit (LWB) line; and a second inverter connected between a global write bit_bar (GWB_bar) line and a local write bit_bar (LWB_bar) line. Each local write driver includes: a third inverter connected between the GWB line and the LWB line; and a fourth inverter connected between the GWB_bar line and the LWB_bar line. In some embodiments, a distributed write driving arrangement, which includes each local write driver being in the corresponding segment of bit cells, has a benefit of mitigating a problem of resistive and/or capacitive loading of the LWB line and the LWB_bar line. In some embodiments, each local write driver is in a first device layer and the global write driver is in a second device layer over the first device layer, which confers a benefit that the global write driver is more easily configured for high speed and a large footprint than a circumstance in which the global write driver is in the first device layer.
(12) FIG. **1** is a block diagram of a semiconductor device **100**, in accordance with at least one embodiment of the present disclosure.

(13) In FIG. **1**, semiconductor device **100** includes, among other things, a circuit macro (hereinafter, macro) **102**. In some embodiments, macro **102** is an SRAM macro. In some embodiments, macro **102** is a macro other than an SRAM macro. Macro **102** includes, among other things, one or more array & column driving regions **104**, each of which includes a distributed write driving arrangement. An example of region **104** is array & column driving region **200** of FIG. **2**.
(14) FIG. **2** is a block diagram of array & column driving region **200** which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure. Region **200** of FIG. **2** is an example of region **104** of FIG. **1**.
(15) In FIG. **2**, region **200** is organized into columns, with columns **207**(*j*) through **207**(*j+n*) being shown in FIG. **2**, where j and n are integers and j≥0, n≥1 and j indicates a column number. Region **200** includes segments **202**A and **202**B, a global driving (g-drv) block **204**; and a control block **206**.
(16) Segment **202**A includes: a block **210**A; a segment-driving (s-drv) block **218**A; and a block **214**A. Segment **202**B includes: a block **210**B; a s-drv block **218**B; and a block **214**B.
(17) Block **210**A is organized as a two-dimensional array of rows and columns, the array including bit cells **212**A(i,j) through **212**A(i+m,j+n), where i and m are integers, i≥0, m≥1, and i indicates a row number. Bit cells, e.g., bit cell **212**A(i,j) are shown in more detail in FIG. **3** (discussed below). For example, bit cells **212**A(i,j) through **212**A(i+m,j) are in column **207**(*j*). Block **214**A is organized as a two-dimensional array including bit cells **216**A(i,j) through **216**A(i+m,j+n). Block **210**B is organized as a two-dimensional array including bit cells **212**B (i,j) through **212**B(i+m,j+n). Block **214**B is organized as a two-dimensional array including bit cells **216**B(i,j) through **216**B (i+m,j+n).
(18) S-drv block **218**A includes local write drivers **220**A(j) through **220**A(j+n) referred to as segment-column (s-col) drivers **220**A(j) through **220**A(j+n). S-col drivers, e.g., s-col driver **220**A(j), are shown in more detail in FIG. **3** (discussed below). For example, s-col driver **220**A(j) is in column **207**(*j*). S-drv block **218**B includes local write drivers **220**B(j) through **220**B(j+n) referred to as s-col drivers **220**B(j) through **220**B(j+n).
(19) Global driving (g-drv) block **204** includes global write drivers **224**(*j*) through **224**(*j+n*) referred to as global-column (g-col) drivers **224**(*j*) through **224**(*j+n*). G-col drivers, e.g., g-col driver **224**(*j*), are shown in more detail in FIG. **3** (discussed below). For example, g-col driver **224**(*j*) is in column **207**(*j*).
(20) In FIG. **2**, because region **200** includes global driving (g-drv) block **204** and s-drv blocks **218**A-**218**B (the latter being correspondingly included in segments **202**A-**202**B), region **200** is regarded as having a distributed write driving arrangement.
(21) Control block **206** includes column driving (c-drv) control units **226**(*j*) through **226**(*j+n*). Control units, e.g., c-drv control unit **226**(*j*), are shown in more detail in FIG. **3** (discussed below). For example, c-drv control unit **226**(*j*) is in column **207**(*j*). C-drv control unit **226**(*j*)-**226**(*j+n*) provide corresponding write-control signals (see FIG. **3**, discussed below).
(22) Region **200** further includes: global write bit (GWB) lines **230**(*j*) through **230**(*j+n*); corresponding global write bit_bar (GWB_bar) lines (not shown, but see FIG. **3** discussed below); local write bit (LWB) lines **234**(*j*) through **234**(*j*+n); and corresponding local write bit_bar (LWB_bar) lines (not shown, but see FIG. **3** discussed below).
(23) In region **200** of FIG. **2**, GWB line **230**(*j*) is connected to each of s-col driver **220**A(j), s-col driver **220**B(j) and g-col driver **224**(*j*). GWB line **230**(*j+n*) is connected to each of s-col driver **220**A(j+n), s-col driver **220**B(j+n) and g-col driver **224**(*j+n*), and the like. LWB line **234**(*j*) is connected to each of bit cells **212**A(i,j)-**212**A(i+m,j), s-col driver **220**A(j), bit cells **216**A(i,j)-**216**A(i+m,j), bit cells **212**B(i,j)-**212**B(i+m,j), s-col driver **220**B(j), bit cells **216**B(i,j)-**216**B(i+m,j) and g-col driver **224**(*j*). LWB line **234**(*j+n*) is connected to each of bit cells **212**A(i,j+n)-**212**A(i+m,j+n), s-col driver **220**A(j+n), bit cells **216**A(i,j+n)-**216**A(i+m,j+n), bit cells **212**B(i,j+n)-**212**B(i+m,j+n), s-col driver **220**B(j+n), bit cells **216**B(i,j+n)-**216**B(i+m,j+n) and g-col

driver **224**(*j*+*n*), and the like.

(24) For simplicity of illustration, region **200** of FIG. **2** has been shown with two segments, **202**A-**202**B. In some embodiments, additional segments are included in region **200**. Also for simplicity of illustration, each of segments, **202**A-**202**B has been shown with one s-drv block, namely corresponding s-drv blocks **218**A-**218**B, such that an intra-segment ratio of blocks of cells (bcell) to s-drv blocks (bsdrv) is bcell:bsdrv=2:1. Other ratios are within the scope of the disclosure. In some embodiments, intra-segment ratios bcell:bsdrv have values other than bcell:bsdrv=2:1.

(25) FIG. **3** is a circuit diagram **300** of an array & column driving region which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure.

(26) Circuit diagram **300** is an example implementation of array & column driving region **200** of FIG. **2**. As such, circuit diagram **300** is an example of region **104** of FIG. **1**.

(27) While circuit diagram **300** of FIG. **3** is more detailed in some respects than the block diagram of region **200** of FIG. **2**, e.g., because the circuit diagram depicts transistors, inverters, NOR gates, a GWB_bar line **332**(*j*); a LWB_bar line **336**(*j*), or the like, the circuit diagram also represents a simplification of the block diagram of region **300**. For the sake of simplicity of illustration, simplifications in circuit diagram **300** include: one column **307**(*j*) rather than multiple columns corresponding to all of columns **207**(*j*)-**207**(*j*+*n*) of region **200**; one segment **302**A rather than two segments corresponding to segments **202**A-**202**B in region **200**; one bit cell **312**A(i,j) in block **310**A rather than multiple bit cells corresponding to all of bit cells **212**A(i,j)-**212**A(i+m,j+n) in block **210**A of segment **202**A of region **200**; one s-col driver **320**A(j) rather than multiple s-col drivers corresponding to s-col drivers **220**A(j)-**220**A(j+n) in s-drv block **218**A in region **200**; one bit cell **316**A(i+m,j) in block **310**B rather than multiple bit cells corresponding to all of bit cells **216**A(i,j)-**216**A(i+m,j+n) in block **214**A of segment **202**A of region **200**; one g-col driver **324**(*j*) rather than multiple g-col drivers corresponding to g-col drivers **224**(*j*)-**224**(*j*+*n*) in region **200**; one c-drv control unit **326**(*j*) rather than multiple c-drv control units corresponding to c-drv control units **226**(*j*)-**226**(*j*+*n*) in region **200**; GWB line **330**(*j*) rather than multiple GWB lines corresponding to GWB lines **230**(*j*)-**230**(*j*+*n*) in region **200**; and LWB line **334**(*j*) rather than multiple LWB lines corresponding to LWB lines **234**(*j*)-**234**(*j*+n) in region **200**.

(28) In FIG. **3**, bit cells **312**A(i,j) and **316**A(i+m,j) are dual port, 8 transistor (8T) SRAM bit cells, with one port representing a write port and one port representing a read port. Other bit cell configurations are within the scope of the disclosure. In some embodiments, bit cells **312**A(i,j) and **316**A(i+m,j) are multi-port SRAM bit cells other than dual port SRAM bit cells. In some embodiments, bit cells **312**A(i,j) and **316**A(i+m,j) are implemented with a number of transistors different than 8 transistors.

(29) In circuit diagram **300**, bit cell **312**A(i,j) includes PMOS transistors P**01**-P**02** and NMOS transistors N**01**-N**06**. Transistors P**01**-P**02** and N**01**-N**02** are configured as an SRAM latch **311**. Transistors N**03** and N**04** represent switches (referred to as pass gates) which selectively connect node **303**A and node_bar **303**B of latch **311** to corresponding LWB line **334**(*j*) and LWB_bar line **336**(*j*). Gate electrodes of transistors N**03** and N**04** are connected to an i.sup.th word write line (WRD) A (AWRD(i)). A signal on AWRD(i) is used to select when transistors N**03** and N**04** are ON/OFF and thereby select when node **303**A and node_bar **303**B of latch **311** is connected to corresponding LWB line **334**(*j*) and LWB_bar line **336**(*j*).

(30) In particular regarding latch **311**, transistors P**01** and N**01** are connected in series between a first reference voltage and a second reference voltage. In some embodiments, the first reference voltage is VDD. In some embodiments, the second reference voltage is VSS. Source and drain electrodes of transistor P**01** are connected correspondingly to VDD and node **303**A. Drain and source electrodes of transistor N**01** are connected correspondingly to node **303**A and VSS. Transistors P**02** and N**02** are connected in series between VDD and VSS. Source and drain electrodes of transistor P**02** are connected correspondingly to VDD and node_bar **303**B. Drain and

source electrodes of transistor N**02** are connected correspondingly to node_bar **303**B and VSS. Gate electrodes of each of transistors P**01** and N**01** are connected to node_bar **303**B. Gate electrodes of each of transistors P**02** and N**02** are connected to node **303**A.

(31) In circuit diagram **300**, bit cell **316**A(i+m,j) includes PMOS transistors P**03**-P**04** and NMOS transistors N**07**-N**12**. Bit cell **316**A(i+m,j) is similar to bit cell **312**A(i,j). For the sake of brevity, the discussion of bit cell **316**A(i+m,j) will focus on differences with respect to bit cell **312**A(i,j).

(32) In bit cell **316**A(i+m,j), transistors P**03**-P**04** and N**07**-N**08** are configured as an SRAM latch **315**. Transistors N**09** and N**10** represent switches (referred to as pass gates) which selectively connect nodes **305**A and **305**B of latch **315** to corresponding LWB line **334**(*j*) and LWB_bar line **336**(*j*) under the control of a signal on line AWRD(i+m).

(33) In FIG. **3**, g-col driver **324**(*j*) includes: inverters **340** and **342**, and an equalizer **325**. Inverter **340** is connected between GWB line **330**(*j*) at a node **360**A and LWB line **334**(*j*) at a node **360**B. Inverter **342** is connected between GWB_bar line **332**(*j*) at a node **362**A and LWB_bar line **336**(*j*) at a node **362**B.

(34) Equalizer **325** is connected between LWB line **334**(*j*) at node **360**B and LWB_bar line **336**(*j*) at node **362**B. Equalizer **325** includes PMOS transistors P**11** and P**12** connected in series between LWB line **334**(*j*) at node **360**B and LWB_bar line **336**(*j*) at node **362**B. Source/drain electrodes of transistor P**11** are connected to node **360**B and a node **363**. Source/drain electrodes of transistor P**12** are connected to node **363** and node **362**B. Gate electrodes of transistors P**11** and P**12** are connected to corresponding nodes **360**A and **362**A.

(35) In FIG. **3**, s-col driver **320**A(j) includes: inverters **344** and **346**, and an equalizer **321**. Inverter **344** is connected between GWB line **330**(*j*) at a node **364**A and LWB line **334**(*j*) at a node **364**B. Inverter **346** is connected between GWB_bar line **332**(*j*) at a node **366**A and LWB_bar line **336**(*j*) at a node **366**B. Inverters **344** and **346**, which are included in s-col driver **320**A(j), accordingly are physically at the interior of segment **302**A.

(36) Equalizer **321** is connected between LWB line **334**(*j*) at node **364**B and LWB_bar line **336**(*j*) at node **366**B. Equalizer **321** includes PMOS transistors P**13** and P**14** connected in series between LWB line **334**(*j*) at node **364**B and LWB_bar line **336**(*j*) at node **366**B. Source/drain electrodes of transistor P**13** are connected to node **364**B and a node **365**. Source/drain electrodes of transistor P**14** are connected to node **365** and node **366**B. Gate electrodes of transistors P**13** and P**14** are connected to corresponding nodes **364**A and **366**A.

(37) In FIG. **3**, c-drv control unit **326**(*j*) includes a NOR gate **348** and a NOR gate **350**. The outputs of NOR gates **348-350** are connected to corresponding nodes **360**B and **362**B. A first input of each of NOR gates **348-350** is connected to a column select_bar (CS_bar) line. A second input of NOR gate **348** is connected to a write data (WD) line. A second input of NOR gate **350** is connected to a write data_bar (WD_bar) line. By using NOR gates **348-350**, c-drv control unit **326**(*j*) reflects an ‘active-low’ configuration. In some embodiments, c-drv control unit **326**(*j*) reflects an ‘active-high’ configuration. In some embodiments in which c-drv control unit **326**(*j*) reflects an active-high configuration, c-drv control unit **326**(*j*) includes corresponding NAND gates in place of NOR gates **348-350**.

(38) In the context of an array and column driving region in an SRAM device according to another approach, and more specifically in the context of one column thereof, it is noted that the other approach does not use a distributed driving arrangement but instead uses a consolidated driving arrangement. As such, the other approach does not include a local write driver in each corresponding segment of bit cells, nor a GWB line, nor a GWB_bar line, and has a consolidated driver (not shown) in place of g-col driver **324**(*j*) and c-drv control unit **326**(*j*). The problem of resistive and/or capacitive loading of the LWB line and the LWB_bar line significantly impairs operation of the arrangement according to the other approach.

(39) For example, according to the other approach, during a write process in which a column is selected and a segment is selected, the LWB line is pre-charged to a logical high value (value H).

After pre-charging, the consolidated driver drives the LWB line with either a value H or a logical low value (value L). Consider a write scenario, according to the other approach, in which the node of the latch (of the bit cell) initially stores value H such that the NMOS transistor connected to the node is turned off because the node_bar stores a corresponding value L, the node of the latch (of the bit cell) is selected to be connected to the LWB line, and the consolidated driver attempts to drive/write the LWB line with a value L. In the write scenario according to the other approach, the NMOS transistor will be turned on and will attempt to pull the LWB line from the pre-charge value H down to value L. The resistive and/or capacitive loading of the LWB line significantly impairs the ability of the corresponding NMOS transistor in the latch of the other approach to pull the WRB line from the pre-charge value H down to value L.

(40) In some embodiments, the distributed write driving arrangement of region **200** has a benefit of mitigating a problem of resistive and/or capacitive loading of LWB line **334**(*j*) and LWB_bar line **336**(*j*). In particular, inverters **344** and **346** are included in s-col driver **320**A(j) and accordingly are at the interior of segment **302**A. Inverters **344** and **346** supplement the driving ability of inverters **340** and **342** of g-col driver **324**(*j*), which mitigates the problem of resistive and/or capacitive loading of LWB line **334**(*j*) and LWB_bar line **336**(*j*). The operation of c-drv control unit **326**(*j*), g-col driver **324**(*j*) and s-col driver **320**A(j) is discussed below in the context of FIGS. **4**A-**4**C.

(41) Circuit diagram **300** of FIG. **3** further includes a local read bit (LRB) line **337**(*j*), a segment read (s-read) circuit **368**(*j*), a global read (g-read) circuit **370**(*j*) and a global read bit (GRB) line **339**(*j*). Also in bit cell **312**A(i,j), transistors N**05** and N**06** are connected as cell-read (c-read) circuit **313**.

(42) Regarding c-read circuit **313**, transistors N**05** and N**06** are connected in series between a local read bit (LRB) line **337**(*j*) and VSS. First and second source/drain electrodes of transistor N**05** are connected to LRB line **337**(*j*) and a node **303**C. First and second source/drain electrodes of transistor N**06** are connected to node **303**C and VSS. The gate electrode of transistor N**05** is connected to an i.sup.th word read bit (WRB) B (BWRD(i)). A signal on BWRD(i) is used to select when transistor N**05** is ON/OFF and thereby select when node **303**C of latch **311** is connected to LRB line **337**(*j*). The gate electrode of transistor N**06** is connected to node_bar **303**B of latch **311**.

(43) In some embodiments, during a read process in which column **307**(*j*) is selected and segment **302**A is selected, LRB line **337**(*j*) is pre-charged to a logical high value (value H). After pre-charging, LRB line **337**(*j*) is connected by s-read circuit **368**(*j*) and g-read circuit **370**(*j*) to GRB line **339**(*j*). Also after pre-charging, the signal on BWRD(i) line is used to turn on transistor N**05**. In a first read scenario, in which node **303**A of latch **311** stores a logical low value (value L)) and node_bar **303**B of latch **311** correspondingly stores a value H, the value H on node_bar **303**B will turn on transistor N**06**. Accordingly, in the first read scenario, transistors N**05** and N**06** together connect LRB line **337**(*j*) to VSS, resulting in LRB line **337**(*j*) taking on a value L, which reflects the value L stored at node **303**A of latch **311**. In a second read scenario, in which node **303**A of latch **311** stores a value H and node_bar **303**B of latch **311** correspondingly stores a value L, the value L on node_bar **303**B will turn off transistor N**06**. Accordingly, in the second read scenario, transistor N**06** prevents LRB line **337**(*j*) from being connected to VSS, resulting in LRB line **337**(*j*) retaining the value H, which reflects the value H stored at node **303**A of latch **311**.

(44) Also in bit cell **316**A(i+m,j), transistors N**11** and N**12** are connected as c-read circuit **317**. C-read circuit **317** is similar to c-read circuit **313**. For the sake of brevity, the discussion of c-read circuit **317** will focus on differences with respect to c-read circuit **313**.

(45) Regarding c-read circuit **313**, first and second source/drain electrodes of transistor N**11** are connected to LRB line **337**(*j*) and a node **305**C. First and second source/drain electrodes of transistor N**12** are connected to node **305**C and VSS. Transistor N**11** selectively connects LRB line **337**(*j*) to node **305**C under the control of a signal on line BWRD(i+m). Transistor N**12** selectively connects node **305**C to VSS under the control of a logical value (L or H) stored at node **305**B of latch **315**.

(46) FIGS. **4**A-**4**C are corresponding circuit diagrams **400**A-**400**C of an array & column driving region which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure. More particularly, each of circuit diagrams **400**A-**400**C shows the same circuit albeit in different control phases of column **407**(*j*) of the array & column driving region.
(47) Each of circuit diagrams **400**A-**400**C is an example implementation of array & column driving region **200** of FIG. **2**. As such, each of circuit diagrams **400**A-**400**C is an example of region **104** of FIG. **1**.
(48) In some respects, each of circuit diagrams **400**A-**400**C is a more detailed version of circuit diagram **300**. For example, each of circuit diagrams **400**A-**400**C shows: inverter **440** as including PMOS transistor P**05** connected and NMOS transistor N**13**, with source/drain electrodes of transistor P**05** connected between VDD and a node **460**B, and source/drain electrodes of transistor N**13** connected between node **460**B and VSS; and inverter **442** as including PMOS transistor P**06** connected and NMOS transistor N**14**, with source/drain electrodes of transistor P**06** connected between VDD and a node **462**B, and source/drain electrodes of transistor N**14** connected between node **462**B and VSS. In some respects, for simplicity of illustration, each of circuit diagrams **400**A-**400**C is a less detailed version of circuit diagram **300**. While each of circuit diagrams **400**A-**400**C shows cells **412**A(i,j) and **416**A(i,j), each of circuit diagrams **400**A-**400**C does not show components included in each of cells **412**A(i,j) and **416**A(i,j).
(49) FIG. **4**A assumes a scenario in which column **407**(*j*) is not selected. FIG. **4**B assumes a scenario in which column **407**(*j*) is selected and in which column **407**(*j*) is being pre-charged before a write operation. FIG. **4**C assumes a scenario in which column **407**(*j*) is selected, and in which data is being written to column **407**(*j*), after column **407**(*j*) has been pre-charged.
(50) Regarding FIG. **4**A (which assumes, again, a scenario in which column **407**(*j*) is not selected), because the inclusion of NOR gates **448-450** in c-drv control unit **426**(*i*) reflects an 'active-low' configuration, column **407**(*j*) is not selected when a signal on line CS_BAR is set to value H. When the signal on line CS_BAR is set to value H, the output of each of NOR gates **448-450** is value L. So long as the signal on line CS_BAR is set to value H, the output of NOR gate **448** at node **460**A will be value H regardless of whether the signal on line WD is set to value L or value H. Accordingly, in FIG. **4**A, the value on signal on line WD is shown as L/H. Similarly, as the signal on line CS_BAR is set to value H, the output of NOR gate **450** at node **462**A will be value H regardless of whether the signal on line WD_bar is set to value L or value H. Accordingly, in FIG. **4**A, the value on signal on line WD_bar is shown as L/H.
(51) Regarding g-col driver **424**(*j*), when the output of NAND gate **448** at node **460**A (which also is the input of inverter **440**) has the value L, then the output of inverter **440** at node **460**B has the value H. More particularly, when the input of inverter **440** at node **460**A has the value L, transistor P**05** is turned on and transistor N**13** is turned off so that, correspondingly, VDD is connected to node **460**A and VSS is disconnected/blocked from node **460**A. Similarly, when the output of NAND gate **450** at node **462**A (which also is the input of inverter **442**) has the value L, then the output of inverter **442** at node **462**B has the value H.
(52) Also regarding g-col driver **424**(*j*), when the outputs of each of NAND gates **448-450** at corresponding nodes **460**A and **462**A have the value L, then equalizer **425** is turned on. More particularly, when the inputs of inverters **440-442** at corresponding nodes **460**A and **462**A have the value L, then corresponding transistors P**11**-P**12** are turned on such that LWB line **434**(*j*) is connected to LWB_bar line **436**(*j*), which facilitates equalizing voltage levels on LWB line **434**(*j*) and LWB_bar line **436**(*j*).
(53) Regarding s-col driver **420**(*j*), when the output of NAND gate **448** at node **460**A (which also is the input at node **464**A of inverter **444** of s-col driver **420**A(j) has the value L, then the output of inverter **444** at node **464**B has the value H. Similarly, when the output of NAND gate **450** at node **462**A (which also is the input at node **466**A of inverter **446** of s-col driver **420**A(j) has the value L,

then the output of inverter **442** at node **462**B has the value H.

(54) Also regarding s-col driver **420**A(j), when the inputs of inverters **444-446** at corresponding nodes **464**A-**466**A have the value L, then equalizer **421** is turned on. More particularly, when the inputs of inverters **444-446** at corresponding nodes **464**A-**466**A have the value L, then corresponding transistors P**13**-P**14** are turned on such that LWB line **434**(*j*) is connected to LWB_bar line **436**(*j*), which facilitates equalizing voltage levels on LWB line **434**(*j*) and LWB_bar line **436**(*j*).

(55) The discussion now turns to FIG. **4**B (which assumes, again, a scenario in which column **407**(*j*) is selected and in which column **407**(*j*) is being pre-charged before a write operation). In FIG. **4**B, a signal on line CS_BAR is set to value L. When the signal on line CS_BAR is set to value L, the output of each of NOR gates **448-450** depends on the values on corresponding lines WD and WD_bar. In FIG. **4**B, each of lines WD and WD_bar has value H. Accordingly, the output of each of NOR gates **448-450** at corresponding nodes **460**A and **462**A is value L.

(56) In FIG. **4**B, regarding g-col driver **424**(*j*), when the output of NAND gate **448** at node **460**A (which also is the input of inverter **440**) has the value L, then the signal propagation is similar to that of FIG. **4**A (discussed above). This is because each of lines WD and WD_bar has value H in FIG. **4**B while line CS_bar has the value L, whereas in FIG. **4**A line CS_bar has the value H (such that it does not matter whether the signals on lines WD and WD_bar are set to value L or value H in FIG. **4**A). Accordingly, in FIG. **4**B, column **407**(*j*) is selected, and LWB line **434**(*j*) and LWB_bar line **436**(*j*) are pre-charged to value H before a write operation occurs.

(57) The discussion now turns to FIG. **4**C (which assumes, again, a scenario in which column **407**(*j*) is selected, and in which data is being written to column **407**(*j*), after column **407**(*j*) has been pre-charged). It is to be recalled that a bit cell, e.g., **412**A(i,j) (but see **312**A(i,j) for greater detail), stores a pair of opposite logical values (L & H or H & L) at the corresponding node pair, e.g., node **303**A & node **303**B. Accordingly, in order to write data to one of the cells in column **407**(*j*), e.g., cell **412**A(i,j), the values which c-drv control unit **426**(*i*) outputs at corresponding nodes **460**A and **462**A are a pair of opposite logical values, either L & H or H & L.

(58) FIG. **4**C is similar to FIG. **4**B except that, in FIG. **4**C, the value on line WD is different than the value on line WD_bar, with the result that the value which c-drv control unit **426**(*i*) outputs at node **460**A is the logical opposite of the value which c-drv control unit **426**(*i*) outputs at node **462**A. Accordingly, in FIG. **4**C, line WD is shown as having value L/H and line WD_bar is shown as having value H/L, and the values which c-drv control unit **426**(*i*) outputs at corresponding nodes **460**A and **462**A are shown as H/L and L/H.

(59) More particularly regarding c-drv control unit **426**(*i*) in FIG. **4**C, a signal on line CS_BAR is set to value L. When the signal on line CS_BAR is set to value L, the output of each of NOR gates **448-450** depends on the values on corresponding lines WD and WD_bar. In FIG. **4**C, line WD has value L/H such that the output of NOR gate **448** at node **460**A is H/L. Line WD_bar has value H/L such that the output of NOR gate **448** at node **462**A is L/H.

(60) Regarding g-col driver **424**(*j*) in FIG. **4**C, when the output of NAND gate **448** at node **460**A (which also is the input of inverter **440**) has the value H/L, then the output of inverter **440** at node **460**B has the value L/H. More particularly, when the input of inverter **440** at node **460**A has the value H, transistor P**05** is turned off and transistor N**13** is turned on so that, correspondingly, VDD is disconnected/blocked from node **460**A and VSS is connected to node **460**A. Alternatively, when the input of inverter **440** at node **460**A has the value L, transistor P**05** is turned on and transistor N**13** is turned off so that, correspondingly, VDD is connected to node **460**A and VSS is disconnected/blocked from node **460**A. Similarly, when the output of NAND gate **450** at node **462**A (which also is the input of inverter **442**) has the value L/H, then the output of inverter **442** at node **462**B has the value H/L.

(61) Also regarding g-col driver **424**(*j*), when the output of NAND gate **448** at node **460**A and the output of NAND gate **450** at node **462**A have different logical values, then equalizer **425** is turned

off. More particularly, because transistors P**11** and P**12** are both PMOS transistors, one of P**11** and P**12** are turned off when corresponding nodes **460**A and node **462**A have different logical values. When the output of NAND gate **448** at node **460**A has the value H/L, then transistor P**11** is turned ON/OFF. When the output of NAND gate **450** at node **462**A has the value L/H, then transistor P**12** is turned OFF/ON. With equalizer **425** turned off, LWB line **434**(*j*) is disconnected/blocked from LWB_bar line **436**(*j*), which prevents equalizing voltage levels on LWB line **434**(*j*) and LWB_bar line **436**(*j*), and thereby facilitates the writing of a pair of opposite logical values (L & H or H & L) in a bit cell of column **407**(*j*), e.g., **412**A(i,j).

(62) Regarding s-col driver **420**A(j), when the output of NAND gate **448** at node **460**A (which also is the input at node **464**A of inverter **444** of s-col driver **420**A(j) has the value H/L, then the output of inverter **444** at node **464**B has the value L/H. Similarly, when the output of NAND gate **450** at node **462**A (which also is the input at node **466**A of inverter **446** of s-col driver **420**A(j) has the value L/H, then the output of inverter **442** at node **462**B has the value H/L.

(63) Equalizer **421** of s-col driver **420**A(j), is similar to equalizer **425**. Accordingly, when the inputs of inverters **444** and **446** at corresponding nodes **464**A and **466**A have the corresponding values H/L and L/H, then equalizer **421** is turned off.

(64) A circuit configuration can be described, e.g., in terms of the relative degrees of optimization of various parameter combinations which the circuit configuration represents. For example, speed is a parameter which represents operational speed of a corresponding circuit. In some embodiments, the relative degrees optimization of speed are referred to as low, moderate and high such that the corresponding circuit is configured to exhibit low speed, moderate speed or high speed, where low<moderate<high. As another example, footprint is a parameter which represents an area consumed/occupied by a corresponding circuit. In some embodiments, the relative degrees optimization of footprint are referred to as small, medium and large such that the corresponding circuit is configured to exhibit a small footprint, a medium footprint or a large footprint, where small<medium<large.

(65) In some embodiments, certain relative optimizations of speed and footprint are referred to as types.

(66) In some embodiments, maximum speed is a parameter representing maximum operational speed (Max Speed) of a corresponding circuit. In some embodiments, footprint is a parameter representing an area consumed by a corresponding circuit. In some embodiments, and as summarized in the following table (Table 1), a Type 1 configuration is configured for a moderate magnitude of Max Speed and a medium footprint, a Type 2 configuration is configured for a low magnitude of Max Speed and a small footprint, and a Type 3 configuration is configured for high magnitude of Max Speed and a large footprint. In some embodiments, the moderate magnitude of Max Speed is about halfway between the low magnitude of Max Speed and the high magnitude of Max Speed. In some embodiments, a difference between the low magnitude of Max Speed and the high magnitude of Max Speed is less than about 30%. In some embodiments, a difference between the small footprint and the large footprint is less than about 30%. In some embodiments, each of the difference between the low magnitude of Max Speed and the high magnitude of Max Speed and the difference between the small footprint and the large footprint is less than about 30%.

(67) TABLE-US-00001 TABLE 1 Type Max Speed Footprint Type-1 moderate medium Type-2 low small Type-3 high large

(68) In FIGS. **3** and **4**A-**4**C, each of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is shown as having the same internal configuration. More particularly, each of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is shown as having the Type-1 configuration. Accordingly, each of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is configured for moderate speed and a medium footprint. In some embodiments, the configuration of one or more of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is different than what is shown in FIGS.

3 and **4**A-**4**C. For example, see FIGS. **5**A-**5**C (discussed below).
(69) FIGS. **5**A-**5**C are corresponding circuit diagrams **500**A-**500**C of array & column driving regions, each of which includes a distributed write driving arrangement, in accordance with corresponding embodiments of the present disclosure.
(70) Each of circuit diagrams **500**A-**500**C is an example implementation of array & column driving region **200** of FIG. **2**. As such, each of circuit diagrams **500**A-**500**C is an example of region **104** of FIG. **1**.
(71) Each of circuit diagrams **500**A-**500**C of corresponding FIGS. **5**A-**5**C is an example variant of circuit diagrams **400**A-**400**C of FIGS. **4**A-**4**C. It is to be recalled that each of circuit diagrams **400**A-**400**C shows the same circuit in different control phases of column **407**(*j*) of the array & column driving region. For purposes of brevity, the discussion of circuits **500**A-**500**C will focus on differences with respect to circuit diagrams **400**A-**400**C of FIGS. **4**A-**4**C.
(72) In FIG. **5**A, s-col driver **520**A(j)" and g-col driver **524**(*j*)""" differ from corresponding s-col driver **420**A(j) and g-col driver **424**(*j*) of FIGS. **4**A-**4**C.
(73) In circuit diagram **500**A, s-col driver **520**A(j)" does not include an equalizer that otherwise would correspond to equalizer **421** of s-col driver **420**A(j). Instead, there is a gap/break **578**A between nodes **564**B and **566**B in s-col driver **420**A(j). In terms of type, s-col driver **520**A(j)" is shown as having the Type-2 configuration, where the double apostrophe (") in reference number **520**A(j)" indicates Type-2. Accordingly, s-col driver **520**A(j) is configured for low speed and a small footprint.
(74) Also in circuit diagram **500**A, g-col driver **524**(*j*)"' includes an equalizer **572**A rather than equalizer **425** of g-col driver **424**(*j*).
(75) Equalizer **572**A is connected between LWB line **534**(*j*) at node **560**B and LWB_bar line **536**(*j*) at node **562**B. Equalizer **572**A includes a PMOS transistor P**51**, a NOR gate **574**A and an inverter **576**A. PMOS transistor P**51** is connected between LWB line **534**(*j*) at node **560**B and LWB_bar line **536**(*j*) at node **562**B. The gate electrode of transistor P**51** is connected to the output of inverter **576**A. The input of inverter **576**A is connected to the output of NOR gate **574**A. The first and second inputs of NOR gate **574**A are connected to corresponding nodes **560**A and **562**A. In terms of type, g-col driver **524**(*j*)"' is shown as having the Type-3 configuration, where the triple apostrophe ("') in reference number **524**(*j*)"' indicates Type-3. Accordingly, g-col driver **524**(*j*)"' is configured for high speed and a large footprint.
(76) In terms of how the values on corresponding nodes **560**A and **562**A turn on/off equalizer **572**A, equalizer **572**A operates the same as equalizer **425** of FIGS. **4**A-**4**C. Because the inclusion of NOR gate **574**A reflects an ‘active-low’ configuration, when each of nodes **560**A and **562**A has value L, then the output of NOR gate **574**A has value H and the gate electrode of transistor P**51** has value L, which turns on transistor P**51**. Any other combination of values on nodes **560**A and **562**A produces results in the output of NOR gate **574**A having value L and the gate electrode of transistor P**51** having value H, which turns off transistor P**51**. Accordingly, in FIG. **5**A, the values on nodes **560**A and **562**A are correspondingly shown as L/H/X/X and L/H/X/X.
(77) FIG. **5**B is similar to FIGS. **4**A-**4**C in some respects and is similar to FIG. **5**A in some respects. In FIG. **5**B, s-col driver **520**A(j) is the same as s-col driver **420**A(j) in FIGS. **4**A-**4**C. Also in FIG. **5**B, g-col driver **524**(*j*)"' is the same as g-col driver **524**(*j*)"' in FIG. **5**A.
(78) FIG. **5**C is similar to FIGS. **4**A-**4**C in some respects and is similar to FIG. **5**A in some respects. In FIG. **5**C, s-col driver **520**A(j)" is the same as s-col driver **520**A(j)" in FIG. **5**A. Also in FIG. **5**C, g-col driver **524**(*j*) is the same as g-col driver **424**(*j*) in FIGS. **4**A-**4**C.
(79) In terms of the types summarized in Table 1, the combinations of types shown in FIGS. **3**, **4**A-**4**C and **5**A-**5**C are summarized in the following table (Table 2).
(80) TABLE-US-00002 TABLE 2 medium small large footprint footprint footprint moderate low high max max max speed speed speed Type 1 Type 2 Type 3 medium moderate max Type 1 FIGS. 3 FIG. 5C s-col footprint speed and 4A-4C driver small low max Type 2 footprint speed large high

max Type 3 FIG. 5B FIG. 5A footprint speed g-col driver
(81) In some embodiments, certain relative optimizations of speed and footprint are summarized in the following table (Table 3).
(82) TABLE-US-00003 TABLE 3 ≈same smaller footprint footprint ≈same lower max speed max speed ≈same ≈same FIGS. 3 and s-col footprint max speed 4A-4C driver larger higher max FIGS. footprint speed 5A-5C g-col driver
(83) Relative to Table 3, in FIGS. **3** and **4**A-**4**C, each of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is configured for substantially the same maximum speed, and each of s-col driver **320**A(j), g-col-driver **324**(*j*), s-col driver **420**A(j) and g-col-driver **424**(*j*) is configured with substantially the same footprint. In FIG. **5**A, s-col driver **520**A(j)" is configured for a lower maximum speed as compared to g-col driver **524**(*j*)"', and s-col driver **520**A(j)" is configured with a smaller footprint as compared to g-col driver **524**(*j*)"'. In FIG. **5**B, s-col driver **520**A(j) is configured for a lower maximum speed as compared to g-col driver **524**(*j*)"', and s-col driver **520**A(j)" is configured with a smaller footprint as compared to g-col driver **524**(*j*)"'. In FIG. **5**C, s-col driver **520**A(j)" is configured for a lower maximum speed as compared to g-col driver **524**(*j*), and s-col driver **520**A(j)" is configured with a smaller footprint as compared to g-col driver **524**(*j*)"'.
(84) FIG. **6** is a cross-section of an array & column driving region **600** which includes a distributed write driving arrangement, in accordance with at least one embodiment of the present disclosure. As such, region **600** of FIG. **6** is an example of region **104** of FIG. **1**. In some embodiments, each of circuit diagrams **300**, **400**A-**400**C and **500**A-**500**C have cross-sections corresponding to FIG. **6**.
(85) FIG. **6** includes layers **671** and **673**. Layer **673** is on layer **671**. Layer **671** is a p.sup.th layer (layer(p)) of devices (not shown), where p is an integer and p≥0. Layer **673** is a (p+1).sup.th layer (layer(p(i+1) of devices (not shown).
(86) Examples of devices included in device layer(p) **671** include: segments **202**A and **202**B of FIG. **2**, which include corresponding s-col drivers **220**A(j)-**220**(*j*+*n*) and **220**B(j)-**220**B(j+n); segment **302**A of FIG. **3**, which includes s-col driver **320**A(j); segment **402**A of FIGS. **4**A-**4**C, which includes s-col driver **420**A(j); segments **502**A″ of FIGS. **5**A and **5**C, which include s-col driver **520**A(j)"; and segment **502**A of FIG. **5**B, which include s-col driver **520**A(j).
(87) Examples of devices included in device layer(p+1) **673** include: g-col drivers **224**(*j*)-**224**(*j*+*n*) and c-drv control units **226**(*j*)-**226**(*j*+*n*) of FIG. **2**; g-col driver **324**(*j*) and c-drv control unit **326**(*j*) of FIG. **3**; g-col driver **424**(*j*) and c-drv control unit **426**(*j*) of FIGS. **4**A-**4**C; g-col driver **524**(*j*)"' of FIGS. **5**A-**5**B; g-col driver **524**(*j*) of FIG. **5**C; and c-drv control unit **526**(*j*) of FIGS. **5**A-**5**C.
(88) Device layer(p) **671** includes sublayers **675**-**679**. Sublayer **677** is on sublayer **675**. Sublayer **679** is on sublayer **677**. Device layer(p+1) **673** includes sublayers **681**-**687**. Sublayer **683** is on sublayer **681**. Sublayer **685** is on sublayer **683**. Sublayer **687** is on sublayer **685**.
(89) Sublayer **679** is a q.sup.th sublayer (sublayer(q)) of metallization, where q is an integer and q≥0. In some embodiments, the q.sup.th sublayer is the first sublayer of metallization, in which case q=0 or q=1 depending upon the numbering convention of the corresponding design rules. Sublayer **683** is a (q+1).sup.th sublayer (sublayer(q+1)) of metallization. In some embodiments, metallization sublayer(q) **679** also includes one or more interconnects (not shown), e.g., one or more vias. In some embodiments, metallization sublayer(q+1) **683** also includes one or more interconnects (not shown), e.g., one or more vias.
(90) Sublayer **675** includes semiconductor structures (not shown), e.g., active regions or the like. Sublayer **677** is an interconnect sublayer which includes interconnects (not shown), e.g., vias. The vias of interconnect sublayer **677** connect semiconductor structures of sublayer **675** to corresponding conductors (not shown) in metallization sublayer(q) **679**. At least some of the devices of layer(p) **671** include one or more semiconductor structures of sublayer **675**, one or more vias of interconnect sublayer **677** and one or more conductors of metallization sublayer(q) **679**.
(91) Sublayer **687** includes semiconductor structures (not shown), e.g., active regions or the like.

Sublayers **681** and **685** are interconnect sublayers, each of which includes interconnects (not shown), e.g., vias. The vias of interconnect sublayer **681** connect conductors (not shown) in metallization sublayer(q+1) to corresponding conductors (not shown) in metallization sublayer(q) **679**. The vias of interconnect sublayer **685** connect semiconductor structures of sublayer **687** to corresponding conductors (not shown) in metallization sublayer(q+1) **683**. At least some of the devices of layer(p+1) **671** include one or more semiconductor structures of sublayer **687**, one or more vias of interconnect sublayer **685** and one or more conductors of metallization sublayer(q) **683**.

(92) In array & column driving region **600** of FIG. **6**, device layer(p+1) **673** is less densely populated than device layer(p) **671**. Accordingly, as contrasted with device layer(p) **671**, device layer(p+1) **673** more easily accommodates circuits with Type-1 configurations (moderate speed and medium footprint) than does device layer(p) **671**, and yet more easily accommodates circuits with Type-3 configurations (high speed and large footprint) than does device layer(p) **671**.

(93) In some embodiments, device layer(p+1) **673** includes Type-1 and/or Type-3 but not Type-2 configurations of circuits, whereas layer(p) **671** includes Type-1 and/or Type-2 but not Type-3 configurations of circuits. In some embodiments, device layer(p+1) **673** includes Type-3 configurations but not Type-1 and/or Type-2 configurations of circuits, whereas layer(p) **671** includes Type-2 configurations but not Type-1 and/or Type-3 configurations of circuits. In some embodiments, device layer(p+1) **673** includes Type-1 configurations but not Type-2 and/or Type-3 configurations of circuits, whereas layer(p) **671** includes Type-1 configurations but not Type-2 and/or Type-3 configurations of circuits. Other configurations are within the scope of the disclosure.

(94) FIG. **7** is a flowchart of a method **700** of write-driving a column in an array & column driving region of a SRAM macro on a distributed basis, in accordance with some embodiments.

(95) Method **700** is implementable, for example, using circuits such as those in FIGS. **4**A-**4**C, **5**A-**5**C, or the like, in accordance with some embodiments.

(96) Regarding method **700**, an example of the SRAM macro is SRAM macro **102**. Examples of the column and the corresponding an array & column driving region include: columns **207**(*j*)-**207**(*j*+*n*) in region **200** of FIG. **2**; column **307**(*j*) in circuit diagram **300** of FIG. **3**; column **407**(*j*) in circuit diagrams **400**A-**400**C of FIGS. **4**A-**4**C; and columns **507**A(j)-**507**C(j) in corresponding circuit diagrams **500**A-**500**C of corresponding FIGS. **5**A-**5**C.

(97) In FIG. **7**, method **700** includes blocks **702-714**. At block **702**, a GWB line is driven with a first signal having a first logical value. An example of the GWB line is GWB line **430**(*j*) in FIGS. **4**A-**4**C. An example of the first signal having the first state is the signal on node **460**A in FIGS. **4**A-**4**B, which has the value L. An example of the first signal having the first state is the signal on node **460**A in FIG. **4**C, which has the value L/H (see FIG. **4**C, discussed above). From block **702** flow proceeds to block **704**.

(98) At block **704**, a GWB_bar line is driven with a second signal which has either the first logical value or a second logical value opposite to the first logical value. An example of the GWB_bar line is GWB_bar line **432**(*j*) in FIGS. **4**A-**4**C. An example of the second signal having the first state is the signal on node **462**A in FIGS. **4**A-**4**B, which has the value L. Another example of the first signal having the second state is the signal on node **462**A in FIG. **4**C, which has the value H/L (see FIG. **4**C, discussed above). From block **704**, flow proceeds to block **706**.

(99) At block **706**, using each of a first inverter in a global write driver and a third inverter in a local write driver, the first signal is inverted to form a first_bar signal having the second logical value. An example of the first inverter in the global write driver is inverter **440** in FIGS. **4**A-**4**C. An example of the third inverter in the local write driver is inverter **444** in FIGS. **4**A-**4**C. An example of the first_bar signal having the second logical value is the signal on node **460**B in FIGS. **4**A-**4**B, which has the value H. Another example of the first signal having the second state is the signal on node **460**B in FIG. **4**C, which has the value L/H (see FIG. **4**C, discussed above). From block **706**,

flow proceeds to block **708**.
(100) At block **708**, using each of a second inverter in a global write driver and a fourth inverter in a local write driver, the second signal is inverted to form a second_bar signal having a logical value opposite the logical value of the second signal. An example of the second inverter in the global write driver is inverter **442** in FIGS. **4**A-**4**C. An example of the fourth inverter in the local write driver is inverter **446** in FIGS. **4**A-**4**C. An example of the second_bar signal is the signal on node **462**B in FIGS. **4**A-**4**B, which has the value H. Another example of the second_bar signal is the signal on node **462**B in FIG. **4**C, which has the value H/L (see FIG. **4**C, discussed above). From block **708**, flow proceeds to block **710**.
(101) At block **710**, the first_bar signal is provided to a first pass gate of each of the bit cells. Examples of the bit cells are bit cells **420**A(i,j) and **416**A(i,j) in FIGS. **4**A-**4**C. An example of the first_bar signal being provided to the first pass gate of a bit cell is the signal on nodes **405**A and **403**A in FIGS. **4**A-**4**C. From block **710**, flow proceeds to block **712**.
(102) At block **712**, the second_bar signal is provided to a second pass gate of each of the bit cells. Again, examples of the bit cells are bit cells **420**A(i,j) and **416**A(i,j) in FIGS. **4**A-**4**C. An example of the second_bar signal being provided to the second pass gate of a bit cell is the signal on nodes **405**B and **403**B in FIGS. **4**A-**4**C. From block **712**, flow proceeds to block **714**.
(103) At block **714**, each of the first and second equalizers is controlled with the first and second signals. Examples of the first and second equalizers are corresponding equalizers **425** and **421** in FIGS. **4**A-**4**C.
(104) In some embodiments, block **714** includes: turning off each of the first equalizer circuit and the second equalizer circuit when the first and second signals have different logical values, an example of which is shown in FIG. **4**C.
(105) In some embodiments, block **714** includes: providing the first signal to gates of a first and third transistor; and providing the second signal to gates of a second and fourth transistor. Examples of the first to fourth transistors are corresponding transistors P**11**-P**14** of FIGS. **4**A-**4**C, the gates of which are connected to corresponding nodes **460**A, **462**A, **464**A and **466**A.
(106) In some embodiments, block **714** includes: logically combining the first and second signals to form a third signal; and providing the third signal to a gate of a first transistor. An example of the transistor is transistor P**51** in FIG. **5**A. An example of logically combining the first and second signals to form a third signal is providing the signals on nodes **560**A and **562**A to NOR gate **574**A and inverting the output of NOR gate **574**A with inverter **576**A, where the third signal is formed at the output of inverter **576**A. An example of providing the third signal to a gate of a first transistor is providing the signal on the output of inverter **576**A to the gate of transistor P**51**.
(107) In some embodiments, a semiconductor memory device includes: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWBL_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled to the segments; and a global write driver coupled to each of the local write drivers.
(108) In some embodiments, the local write driver is in a first device layer of the semiconductor device; and the global write driver is in a second device layer over the first device layer of the semiconductor device.
(109) In some embodiments, each the local write drivers is at an interior location in the corresponding segment; the bit cells are in the first device layer; the LWB line and the LWB_bar line are in a first metallization layer, the first metallization layer being between the first device layer and the second device layer; and the GWB line and the GWBL_bar line are in a second metallization layer, the second metallization layer being between the first metallization layer and the second device layer.
(110) In some embodiments, the global write driver includes: a global equalizer circuit arranged in

a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
(111) In some embodiments, the global equalizer circuit includes first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
(112) In some embodiments, each of the local write drivers includes: a segment equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
(113) In some embodiments, the segment equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
(114) In some embodiments, the global write driver includes a first buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line.
(115) In some embodiments, the first buffer includes a first inverter and the second buffer includes a second inverter.
(116) In some embodiments, each of the local write drivers includes a first buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line.
(117) In some embodiments, the first buffer includes a first inverter and the second buffer includes a second inverter.
(118) In some embodiments, a semiconductor memory device includes: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWBL_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled such that, for each of the segments, the corresponding local write driver being coupled between the local write bit (LWB) line and the local write bit_bar (LWB_bar) line; and a global write driver coupled to each of the local write drivers.
(119) In some embodiments, the local write driver is in a first device layer of the semiconductor device; and the global write driver is in a second device layer over the first device layer of the semiconductor device.
(120) In some embodiments, each the local write drivers is at an interior location in the corresponding segment; the bit cells are in the first device layer; the LWB line and the LWB_bar line are in a first metallization layer, the first metallization layer being between the first device layer and the second device layer; and the GWB line and the GWBL_bar line are in a second metallization layer, the second metallization layer being between the first metallization layer and the second device layer.
(121) In some embodiments, the global write driver includes: a global equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
(122) In some embodiments, the global equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
(123) In some embodiments, a semiconductor memory device includes: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWBL_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled to the segments, each local write driver including a first buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line; and a global write driver coupled

to each of the local write drivers.
(124) In some embodiments, the first buffer includes a first inverter and the second buffer includes a second inverter.
(125) In some embodiments, each of the local write drivers includes: a segment equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
(126) In some embodiments, the segment equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
(127) It will be readily seen by one of ordinary skill in the art that one or more of the disclosed embodiments fulfill one or more of the advantages set forth above. After reading the foregoing specification, one of ordinary skill will be able to affect various changes, substitutions of equivalents and various other embodiments as broadly disclosed herein. It is therefore intended that the protection granted hereon be limited only by the definition contained in the appended claims and equivalents thereof.

Claims

1. A semiconductor memory device comprising: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWB_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled to the segments; and a global write driver coupled to each of the local write drivers.
2. The semiconductor memory device of claim 1, wherein: the local write driver is in a first device layer of the semiconductor device; and the global write driver is in a second device layer over the first device layer of the semiconductor device.
3. The semiconductor memory device of claim 2, wherein: each the local write drivers is at an interior location in the corresponding segment; the bit cells are in the first device layer; the LWB line and the LWB_bar line are in a first metallization layer, the first metallization layer being between the first device layer and the second device layer; and the GWB line and the GWB_bar line are in a second metallization layer, the second metallization layer being between the first metallization layer and the second device layer.
4. The semiconductor memory device of claim 1, wherein the global write driver includes: a global equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
5. The semiconductor memory device of claim 4, wherein: the global equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
6. The semiconductor memory device of claim 1, wherein each of the local write drivers includes: a segment equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.
7. The semiconductor memory device of claim 6, wherein: the segment equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.
8. The semiconductor memory device of claim 1, wherein: the global write driver includes a first

buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line.

9. The semiconductor memory device of claim 8, wherein: the first buffer includes a first inverter and the second buffer includes a second inverter.

10. The semiconductor memory device of claim 1, wherein: each of the local write drivers includes a first buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line.

11. The semiconductor memory device of claim 10, wherein: the first buffer includes a first inverter and the second buffer includes a second inverter.

12. A semiconductor memory device comprising: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWB_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled such that, for each of the segments, the corresponding local write driver being coupled between the LWB line and the LWB_bar line; and a global write driver coupled to each of the local write drivers.

13. The semiconductor memory device of claim 12, wherein: the local write driver is in a first device layer of the semiconductor device; and the global write driver is in a second device layer over the first device layer of the semiconductor device.

14. The semiconductor memory device of claim 13, wherein: each the local write drivers is at an interior location in the corresponding segment; the bit cells are in the first device layer; the LWB line and the LWB_bar line are in a first metallization layer, the first metallization layer being between the first device layer and the second device layer; and the GWB line and the GWB_bar line are in a second metallization layer, the second metallization layer being between the first metallization layer and the second device layer.

15. The semiconductor memory device of claim 12, wherein the global write driver includes: a global equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.

16. The semiconductor memory device of claim 15, wherein: the global equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.

17. A semiconductor memory device comprising: a local write bit (LWB) line; a local write bit_bar (LWB_bar) line; a global write bit (GWB) line; a global write bit_bar (GWB_bar) line; a column of segments, each segment including bit cells; each of the bit cells being coupled correspondingly between the LWB and LWB_bar lines; and a distributed write driving arrangement including: local write drivers correspondingly coupled to the segments, each local write driver including a first buffer coupled between the LWB line and the GWB line and a second buffer coupled between the LWB line and the GWB line; and a global write driver coupled to each of the local write drivers.

18. The semiconductor memory device of claim 17, wherein: the first buffer includes a first inverter and the second buffer includes a second inverter.

19. The semiconductor memory device of claim 17, wherein each of the local write drivers includes: a segment equalizer circuit arranged in a switched-coupling between the LWB line and the LWB_bar line, and arranged in a control-coupling with respect to signals correspondingly on the GWB line and the GWB_bar line.

20. The semiconductor memory device of claim 19, wherein: the segment equalizer circuit includes: first and second transistors coupled in series between the LWB line and the LWB_bar line; and gate electrodes of the first and second transistors are coupled correspondingly to the GWB line and the GWB_bar line.